DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

INTERNSHIP REPORT

ON

"Aviation-Specific Turbulence Forecasting using AI-ML Techniques"

Submitted in partial fulfillment for the award of degree(20INT82)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Suprith A.S - 1DS21AI052

Conducted at

CSIR-National Aerospace Laboratories (NAL)

Department of Artificial Intelligence and Machine Learning

Dayananda Sagar College of Engineering Bangelore-78

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

CERTIFICATE

This is to certify that the Internship titled **"Aviation-Specific Turbulence Forecasting using AI-ML Techniques"** carried out by **Mr. Suprith A.S [1DS21AI052]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)

______________	______________	______________
Signature of Reporting Head	**Signature of Internship Coordinator**	**Signature of HoD**
Dr. Mrudula G	**Prof. Ensteih Silvia**	**Dr. Vindhya P Malagi**
Senior Principal Scientist	Dept. of AI & ML	Dept. of AI & ML
CEM	DSCE	DSCE
CSIR-NAL		

External Viva:

Name of the Examiner	Signature with Date
1) ___________________	________________
2) ___________________	_________________

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

D E C L A R A T I O N

I, **Suprith A.S,** final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **CSIR-NAL**. This report is submitted in partial fulfillment of the requirements for the award of bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date: USN: 1DS21AI052

Place: Bangalore NAME: Suprith A.S

OFFER LETTER

सीएसआईआर-राष्ट्रीय वांतरिक्ष प्रयोगशालाएं
CSIR - National Aerospace Laboratories
PB No 1779, HAL Airport Road, Kodihalli, Bengaluru-560 017, India
080-2508 6250 +91-94810 54732
head dts@nal.res.in (80)-2508 6009 www.nal.res.in

CSIR-NAL
AS 9100:2016
Certified

डॉ. एस टी अरुणा
Dr. S T Aruna
Chief Scientist & Head
Directors Technical Secretariat

27th February, 2025

Dr. Vindhya P Malagi
Head of Dept of AI & Machine Learning
Dayananda Sagar College
Bengaluru

Email: hod-ai@dayanandasagar.edu; divyas@nal.res.in;

Sub: Internship Work at NAL

Dear Sir,

This has reference to your letter dated 05.02.2025 on the above subject. CSIR-NAL provides internship to the students so that they get exposed to the latest technologies available in CSIR-NAL in the field of Aeronautical and Aerospace related activities. This will help to mould their future. We provide internship to the students on NO PAY NO FEE basis.

We are pleased to inform you that the Director, CSIR-NAL has kindly approved to provide an opportunity to your **B.E AI student, Mr. Suprith A S** for carrying out the internship work/Project work at NAL. The student will be working under the guidance of **Ms. Divya, SPS, NTAF.**

Kindly advise your student to meet the undersigned on any working days (except Saturdays and Sundays) along with this letter and a photo copy of College ID Card at the time of admission.

Please find enclosed the authentication form to be duly attested by the Principal and should be submitted at the time of joining NAL.

This is for your kind information.

Thanking you,

Yours sincerely,

S.T.A.

Head, DTS.

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. I take this opportunity to express my gratitude to all who have helped me to complete the Internship.

I express my sincere thanks to the Principal **Dr. B G Prasad**, for providing me adequate facilities to undertake this Internship.

I would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing me an opportunity to carry out Internship and for her valuable guidance and support.

I would like to thank **Dr. Mrudula G**, Senior Principal Scientist in Centre for Electromagnetics, CSIR-NAL for guiding me during the period of internship.

I express my deep and profound gratitude to my Guide, **Prof. Ensteih Silvia,** Associate Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

I would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

I would like to thank the non-teaching members of the Department of AI & ML, for helping me during the Internship.

Last but not the least, I would like to thank my parents and friends without whose constant help, the completion of Internship would have not been possible.

Suprith A.S [1DS21AI052]

TABLE OF CONTENTS

CHAPTER 1
INTRODUCTION

INTRODUCTION TO AVIATION-SPECIFIC TURBULENCE FORECASTING USING AI-ML TECHNIQUES

This project, titled Aviation-Specific Turbulence Forecasting using AI-ML Techniques represents a significant stride toward enhancing flight safety and efficiency by leveraging advanced techniques in Artificial Intelligence and Machine Learning. It focuses on integrating diverse atmospheric datasets to develop robust turbulence detection mechanisms. The initiative is anchored in state-of-the-art research and practical applications, offering innovative solutions to predict turbulent conditions by analysing upper air sounding data and other key meteorological parameters.

At the heart of the project is the utilization of sophisticated Python-based tools designed to extract, process, and analyse atmospheric data. This technical framework not only harnesses machine learning algorithms for pattern recognition in wind directions and other relevant variables but also ensures that insights are actionable for real-time applications in aviation. By automating the extraction of upper air sounding data, the project aims to facilitate early detection of anomalies that could indicate the onset of turbulence, thus contributing to safer flight operations.

further underscores the practical relevance and industry significance of this work. With a rich history of blending academic rigor with practical innovation, CSIR-NAL provides an excellent environment for implementing research ideas into tangible solutions. This internship project benefits immensely from this association, enabling the integration of academic insights with field-tested methodologies, and ensuring that the outcomes are both scientifically sound and operationally viable.

This project not only advances the current state of atmospheric data utilization for turbulence detection but also highlights the strategic role of machine learning in modern aerospace applications. It sets the stage for future developments in aviation safety and operational efficiency, establishing a model framework that could be adopted widely across industry. The report delves further into the technical processes, challenges encountered, and lessons learned, providing a comprehensive overview of the project's contributions to the field.

Weather describes the short-term state of the atmosphere at a particular place, encompassing phenomena like temperature, humidity, precipitation, air pressure, and wind. It is primarily driven by differences in air pressure, temperature, and moisture, which arise from factors like the sun's angle at varying latitudes and the physical characteristics of different surfaces. Understanding weather is crucial for various aspects of human life, from daily activities to long-term planning, and it is measured using a range of instruments and sophisticated technologies.

Aviation weather focuses on atmospheric conditions critical for safe and efficient flight operations. Several factors are particularly important:

- **Convective Weather (CW):** This includes thunderstorms, with their associated updrafts, downdrafts, lightning, hail, and heavy rain. These phenomena create highly turbulent conditions and pose significant hazards to aircraft.

- **Windshear:** Defined as a sudden change in wind velocity (speed or direction) over a short distance, especially with height. Low-level wind shear is a major hazard during takeoff and landing, potentially causing rapid changes in airspeed and lift, leading to dangerous situations. Strong wind shear can also occur at higher altitudes, sometimes leading to clear-air turbulence.

- **Turbulence:** Irregular motion of the air caused by eddies and vertical currents. It can range from light bumps to severe or extreme, causing abrupt changes in aircraft attitude and height, and even temporary loss of control. Turbulence can be caused by mechanical friction with terrain, convection (e.g., in thunderstorms), mountain waves, and wind shear instabilities (Clear Air Turbulence - CAT).

Other Important Factors: These include icing, visibility (fog, haze), cloud ceilings, volcanic ash, and strong winds that can affect aircraft drift and fuel consumption. Aviation weather advisories like SIGMETs and AIRMETs are issued to alert pilots to potentially hazardous conditions.

Weather observations involve collecting data on atmospheric conditions using various tools and systems. These measurements provide the initial conditions for weather forecasting. Key instruments include:

1. **Thermometers:** Measure air temperature.

2. **Barometers:** Measure atmospheric pressure. Changes in pressure are crucial indicators of approaching weather systems.

3. **Anemometers:** Measure wind speed, often integrated with wind vanes to also show wind direction.

4. **Hygrometers:** Measure humidity levels.

5. **Rain gauges:** Measure the amount of liquid precipitation.

6. **Ceilometers:** Measure cloud ceiling (height of the lowest cloud layer).

7. **Transmissometers:** Measure visibility.

8. **Weather balloons (Radiosondes):** Carried aloft to measure temperature, humidity, pressure, and wind at various altitudes, providing crucial upper-air data.

9. **Doppler Radar:** Detects precipitation, its intensity, and movement, and can also detect wind shear within storms.

10. **Weather Satellites:** Provide continuous, wide-area surveillance of cloud patterns, temperature, water vapor, and track the movement of weather systems globally.

11. **Automated Surface Observing Systems (ASOS) and Remote Automated Weather Stations (RAWS):** Provide continuous, automated measurements of surface weather variables.

12. **Aircraft sensors:** Aircraft themselves report atmospheric data, including temperature, wind, and even turbulence encountered (PIREPs - Pilot Weather Reports).

Forecasting Models and Techniques: Modern weather forecasting primarily relies on Numerical Weather Prediction (NWP) models. These models use complex mathematical equations based on the laws of physics to simulate atmospheric processes and predict future weather states. They take vast amounts of observed data as initial conditions and run sophisticated simulations on high-performance computers.

Current techniques involve:

- **Deterministic Models:** Produce a single forecast based on the most likely atmospheric evolution.
- **Ensemble Prediction Systems (EPS):** Run multiple simulations with slightly varied initial conditions or model physics to account for inherent uncertainties in the atmosphere. This provides a range of possible outcomes and helps assess the confidence or probability of a particular weather event.
- **Nowcasting:** Short-range forecasting (0-6 hours) that uses real-time radar, satellite, and observational data to predict small-scale features like thunderstorms with high accuracy.
- **Statistical Methods:** Still used, sometimes in conjunction with NWP, often for localized or specific variable predictions, or to refine model outputs.

The performance of these models has steadily improved over decades, leading to more accurate forecasts for general weather variables like temperature, precipitation, wind, and pressure patterns. Advancements in computing power, data assimilation techniques, and model resolution have significantly reduced forecast errors and extended the useful range of predictions. However, challenges remain in accurately predicting rapidly evolving small-scale phenomena.

Current Status of Turbulence Forecasting: Turbulence forecasting remains a significant challenge due to its transient, complex, and often small-scale nature. It is hard to predict accurately because:

- **Scale Mismatch:** Turbulence often occurs at scales smaller than what current NWP models can explicitly resolve, requiring the use of parametrization (simplified representations) which introduce uncertainties.
- **Complex Interactions:** It arises from various interacting factors, including wind shear instabilities, convection, and mountain waves, making it difficult to isolate and predict individual mechanisms.

- **Limited Observations:** Direct observations of turbulence in the atmosphere are sparse, primarily relying on pilot reports (PIREPs) which are subjective and not always real-time or widespread enough to provide a complete picture. Clear air turbulence (CAT), in particular, is difficult to detect directly as it lacks visible indicators.

- **Chaotic Nature:** The atmosphere is inherently chaotic, and turbulence is a manifestation of this chaos, making precise prediction inherently difficult.

Currently, turbulence forecasting relies on:

- **NWP Model Diagnostics:** Meteorologists use various diagnostic tools and indices derived from NWP model outputs that indicate regions where conditions are favorable for turbulence development (e.g., areas of strong wind shear, atmospheric instability).

- **Pilot Reports (PIREPs):** These real-time observations from pilots are crucial for verifying forecasts and identifying areas of actual turbulence, often feeding back into operational guidance.

- **Ensemble Forecasting:** Running multiple model simulations helps assess the probability and intensity of turbulence, providing forecasters with a range of possible outcomes.

- **Machine Learning/Statistical Approaches:** These are increasingly being used to identify patterns in historical data (NWP outputs, PIREPs, remote sensing) to predict turbulence.

Improvements for the future:

- **Higher Resolution Models:** Developing NWP models with finer spatial and temporal resolution will allow for better explicit representation of turbulence-generating mechanisms.

- **Improved Remote Sensing:** Advancements in remote sensing technologies (e.g., LIDAR, advanced radar) that can directly detect turbulence or its precursors will provide more accurate real-time observations.

- **Enhanced Data Assimilation:** Better integration of various observational data sources, including aircraft reports and satellite data, into models.

- **Better Understanding of Turbulence Dynamics:** Continued research into the fundamental physical processes that generate and sustain turbulence.

- **Community-wide Data Sharing:** More robust systems for sharing turbulence observations and forecasts among airlines and meteorological agencies.

AI/ML in Weather Forecasting:

Artificial intelligence (AI) and machine learning (ML) offer significant advantages over traditional methods in weather forecasting, particularly for complex phenomena like turbulence:

- **Processing Vast Datasets:** AI/ML models can rapidly process and learn from enormous, high-dimensional datasets (historical weather data, satellite imagery, radar data, sensor readings), identifying complex, non-linear patterns that traditional physics-based models or human analysis might miss.

- **Faster Forecasts:** AI models can generate forecasts much faster than traditional NWP models, sometimes in minutes, enabling more frequent updates and improved nowcasting for rapidly evolving conditions.

- **Improved Accuracy:** ML models can learn from past forecast errors and continuously refine their algorithms, leading to potentially higher accuracy, especially for localized predictions and extreme events. They can also excel at predicting "edge cases" or unusual weather patterns.

- **Better Handling of Uncertainty:** ML can be used to develop probabilistic forecasts, providing insights into the likelihood of different outcomes, which is vital for decision-making in aviation.

- **Integration of Diverse Data:** AI/ML can seamlessly integrate diverse data types, including observations from non-traditional sources (e.g., crowd-sourced weather data, IoT sensors), improving the overall understanding of the atmosphere.

- **Direct Turbulence Prediction:** AI/ML can be trained directly on past turbulence events (using PIREPs, for instance) and atmospheric variables to predict future turbulence occurrences and intensity more precisely, without relying solely on physics-based diagnostics. This is particularly beneficial for CAT.

- **Personalized Forecasts:** AI can generate highly localized and personalized forecasts, crucial for specific flight routes or airport operations.

In the turbulence forecasting, AI/ML can:

- **Learn from PIREPs:** By analyzing vast archives of pilot reports in conjunction with atmospheric data, ML models can identify atmospheric conditions that consistently lead to reported turbulence.

- **Leverage Satellite and Radar Data:** AI can extract subtle signatures of turbulence-generating phenomena from high-resolution satellite imagery and radar data.

- **Combine Multiple Predictors:** Instead of relying on individual diagnostic indices, AI can learn complex interactions between various atmospheric variables that contribute to turbulence.

- **Provide Probabilistic Forecasts:** ML models can quantify the probability of different turbulence intensities, giving pilots and dispatchers better risk assessment tools.

While traditional NWP models provide the foundational understanding of atmospheric physics, AI/ML acts as a powerful complement, excelling in pattern recognition, data synthesis, and rapid prediction, especially for challenging phenomena like turbulence, ultimately leading to safer and more efficient air travel.

CHAPTER 2
DESCRIPTION OF ORGANIZATION

NATIONAL AEROSPACE LABORATORIES (CSIR-NAL)

CSIR-NAL

ABOUT THE COMPANY

The Council of Scientific and Industrial Research – National Aerospace Laboratories (CSIR-NAL) is India's premier aerospace research and development organization, established in 1959. Located in Bengaluru, Karnataka, CSIR-NAL operates under the aegis of the CSIR, which is the country's largest research and development organization. NAL's creation was a strategic move to build indigenous capabilities in aeronautical sciences and technologies, aligning with the nation's ambition to become self-reliant in the aerospace sector.

CSIR-NAL is renowned for its wide-ranging expertise in aeronautics, spanning disciplines such as aerodynamics, avionics, aerospace materials, structural design, and propulsion systems. Over the decades, the organization has carved out a significant role in both fundamental and applied research, often acting as a bridge between academic knowledge and industrial application. Its multi-disciplinary R&D capabilities make it a crucial contributor to both civilian and military aviation in India.

One of the key mandates of CSIR-NAL is to develop indigenous technologies to support India's aerospace sector, thereby reducing dependence on imports. The organization has played a pioneering role in the development of small civilian aircraft like the HANSA and SARAS. The SARAS is the first Indian multi-purpose civilian aircraft in the light transport category.

CSIR-NAL's infrastructure includes state-of-the-art laboratories, wind tunnels, computational fluid dynamics (CFD) facilities, and testing equipment, making it one of the most advanced aerospace research centers in Asia. These facilities have enabled the organization to provide vital testing and certification services to various aerospace manufacturers and government agencies. Its work supports critical programs run by entities such as Hindustan Aeronautics Limited (HAL), the Indian Space Research Organisation (ISRO), and the Defence Research and Development Organisation (DRDO).

VISION:

To be a world-class aerodynamic testing facility that propels innovation in aerospace design by enabling breakthrough experiments in multi-regime flight dynamics.

MISSION:

To deliver precise aerodynamic data and comprehensive experimental support through cutting-edge wind tunnel technologies, fostering advancements in aerospace configurations and ensuring high performance, efficiency, and safety of future aerospace systems.

CORE AREAS OF INNOVATION:

- **Civil Aircraft Design and Development:** NAL focuses on designing and building both small and medium-sized civil aircraft.

- **Micro Aerial Vehicle (MAV) Design:** They develop and test technologies for MAVs, including autopilot systems and vision systems.

- **Aerodynamics and Flight Dynamics:** NAL conducts research in both computational and experimental aerodynamics, as well as flight mechanics and control.

- **Aerospace Materials and Structures:** Their research extends to composites, structural design, analysis, and testing, and surface modification of aerospace materials.

- **Aerospace Electronics and Systems:** NAL develops avionics and embedded systems, including technologies for autopilot hardware and vision systems for MAVs.

- **Propulsion Systems:** They research and develop technologies related to turbo machinery, combustion, and propulsion systems.

- **Electromagnetics:** NAL's research in electromagnetics includes radar cross-section studies, radar absorbing material (RAM) technology, and radome technologies.

- **Meteorological Modeling and Wind Energy:** NAL also conducts research in these areas, contributing to various national programs.

- **National Trisonic Aerodynamic Facilities:** They operate state-of-the-art wind tunnels, including the 1.2m Trisonic wind tunnel, which is crucial for testing and characterizing high-speed flight vehicles

CHAPTER 3
SCOPE OF WORK

1. Identify Variables Affecting Turbulence:

Turbulence in the atmosphere is a complex phenomenon driven by various factors. The primary variables influencing its formation and intensity include:

- **Wind Shear:** Changes in wind speed or direction over a short distance, both horizontally and vertically. This is a primary driver of clear-air turbulence (CAT) and low-level wind shear.
- **Atmospheric Stability/Instability:** The tendency of the atmosphere to resist or enhance vertical motion. Unstable conditions (e.g., strong convection) lead to turbulent updrafts and downdrafts.
- **Terrain/Orographic Effects:** The interaction of wind with mountainous terrain creates mountain waves and mechanical turbulence.
- **Convective Activity:** Thunderstorms and deep convection are inherently turbulent due to strong updrafts, downdrafts, and associated wind shear.
- **Jet Streams:** Strong wind shear often exists at the edges of jet streams, contributing to CAT.
- **Frontal Systems:** Areas of significant temperature and wind gradients associated with weather fronts can generate turbulence.
- **Cloud Properties:** The presence and type of clouds can be indicative of atmospheric processes that lead to turbulence. For instance, cumulonimbus clouds are highly turbulent.

Atmospheric turbulence is influenced by several key factors. Wind shear, or abrupt changes in wind speed or direction, is a primary driver, particularly for clear-air turbulence. Additionally, atmospheric stability, terrain interactions, convective activity like thunderstorms, and the presence of jet streams and frontal systems all significantly contribute to the formation and intensity of turbulence.

2. Identify Datasets Containing Relevant Information:

To train and validate turbulence prediction models, access to comprehensive meteorological datasets is crucial. Potential sources include:

- **Copernicus Climate Change Service (CDS) / ECMWF Reanalysis Data (ERA5, ERA5-Land):** These datasets provide global, gridded atmospheric reanalysis data, including a wide range of atmospheric variables at various pressure levels and resolutions. ERA5, in particular, is highly valuable due to its high resolution and comprehensive variable list.
- **Indian Monsoon Data Assimilation and Analysis (IMDAA):** Operated by the India Meteorological Department (IMD), IMDAA provides high-resolution reanalysis data specifically for the Indian subcontinent, which would be crucial for localized studies in that region.
- **Upper Air Sounding Data (Radiosondes):** Data collected from weather balloons provide direct, high-resolution vertical profiles of temperature, humidity, pressure, and wind speed/direction. This is invaluable for understanding vertical atmospheric structure.

3. Selection of Exact Variables for Turbulence Predictions:

Based on the understanding of turbulence drivers and initial data exploration, the following atmospheric variables are highly relevant and will be considered for turbulence prediction:

- **Temperature (t):** Essential for determining atmospheric stability and identifying temperature gradients associated with fronts and jet streams.
- **U-component of Wind (u):** wind component, critical for calculating wind shear and vorticity.
- **V-component of Wind (v)**: wind component, also critical for calculating wind shear and vorticity.
- **Vertical Velocity (w):** Indicates upward or downward air movement, directly related to convection and mountain waves, and a key indicator of vertical shear.
- **Vorticity (relative) (vo):** A measure of the local rotation of the fluid (atmosphere). High relative vorticity often indicates areas of strong shear and potential turbulence.

4. Using GrADS for Data Visualization and Initial Insights:

GrADS (Grid Analysis and Display System) will be utilized for visualizing NetCDF (Network Common Data Form) data. This is a powerful tool for:

- **Data Exploration:** Quickly plotting various atmospheric variables on different pressure levels or altitudes.
- **Spatial Analysis:** Visualizing the distribution of selected variables over specific geographic regions.
- **Temporal Analysis:** Examining the evolution of these variables over time.
- **Identification of Key Features:** Visually identifying regions of high wind shear, strong vertical motion, or high vorticity that might correspond to turbulence.

This visual exploration will help in understanding the relationships between the selected variables and atmospheric conditions conducive to turbulence.

5. Conclusion from GrADS Analysis and Final Variable Selection

Through the GrADS analysis of various variables, including:

- Divergence, Fraction of cloud cover, Geo potential, Ozone mass mixing ratio, Potential vorticity, U-component of wind, V-component of wind, Vertical velocity, Vorticity (relative), Specific cloud ice water content, Specific cloud liquid water content, Specific humidity, Specific rain water content, Specific snow water content, Relative humidity, Temperature.

It has been concluded that the following core variables are most promising for direct input into future turbulence prediction models due to their strong physical correlation with turbulence mechanisms and their widespread availability in reanalysis datasets:

- **Temperature (t), U-component of wind (u), V-component of wind (v), Vertical velocity (w), Vorticity (relative) (vo)**

These variables directly capture wind shear and wind speed (from u and v), which are fundamental to turbulence generation.

Turbulence Prediction Models Using AI/ML

The following AI/ML model architectures will be investigated for turbulence prediction:

1. LSTM + CNN (Long Short-Term Memory + Convolutional Neural Network):

- This hybrid architecture is well-suited for spatio-temporal data, which is characteristic of atmospheric phenomena.
- **CNN (Convolutional Neural Network):** Excellent at learning spatial features and patterns from gridded data (e.g., temperature fields, wind fields). It can capture relationships between neighboring grid points that indicate shear or localized atmospheric features.
- **LSTM (Long Short-Term Memory):** Highly effective for processing sequential data and capturing temporal dependencies. This is crucial for forecasting, as atmospheric conditions evolve over time.
- **Approach:** CNN layers would process the spatial features of the selected atmospheric variables at each time step. The outputs of the CNN would then be fed into LSTM layers to learn the temporal evolution of these features and make future predictions of turbulence. This model can predict turbulence intensity or probability at specific locations and future times.

2. PINN with RANS (Physics-Informed Neural Networks with Reynolds-Averaged Navier-Stokes):

- PINNs are a powerful paradigm that integrates physical laws directly into the neural network's loss function, ensuring that the predictions are physically consistent. RANS equations are commonly used in fluid dynamics to model turbulent flows by averaging the Navier-Stokes equations.
- **Approach:** Instead of purely data-driven learning, the PINN would be designed such that its predictions of turbulence-related fields (e.g., turbulent kinetic energy, Reynolds stresses) adhere to the RANS equations. The neural network would learn the closure models for the RANS equations (which are often empirical) directly from data while being constrained by the fundamental physics.
- This approach could be particularly beneficial for resolving sub-grid scale turbulence or improving parametrization within larger-scale models.
- This might be more computationally intensive and require a strong understanding of fluid dynamics and numerical methods for turbulence.

3. RF + XGBoost (Random Forest + Extreme Gradient Boosting):

- These are powerful ensemble learning algorithms known for their accuracy, robustness to noisy data, and ability to handle high-dimensional datasets. They are relatively less computationally demanding than deep learning models and can provide good baseline performance.
- **Random Forest (RF):** Builds multiple decision trees and averages their predictions. It's good at handling non-linear relationships and is less prone to overfitting.
- **XGBoost (Extreme Gradient Boosting):** An optimized distributed gradient boosting library designed to be highly efficient, flexible, and portable. It excels at structured/tabular data and often achieves state-of-the-art results in classification and regression tasks.
- **Approach:** From the selected variables (t,u,v,w,vo), various turbulence indices and derived features (e.g., vertical wind shear, horizontal wind shear, Richardson number, specific turbulence diagnostic indices) would be engineered.
- **Training:** Both RF and XGBoost models would be trained on these engineered features, with the target variable being a measure of turbulence (e.g., discrete turbulence intensity levels based on PIREPs, or a continuous turbulence metric if available).
- **Ensemble/Comparison:** The performance of both models would be compared, and potentially, an ensemble of their predictions could be used to improve robustness.

This comprehensive plan covers data acquisition, essential variable selection, initial exploratory analysis, and the selection of advanced AI/ML models to build robust turbulence prediction capabilities.

CHAPTER 4
LEARNING OBJECTIVES

1. Exposure to Research at NAL in CEM (Centre for Electromagnetics)

The National Aerospace Laboratories (NAL) in India is a key place for studying aerospace topics, and its Centre for Electromagnetics (CEM) focuses on how turbulence affects things like aircraft safety. CEM combines aerodynamics—the study of air movement—with electromagnetics, which involves how electromagnetic waves behave. This research helps figure out how turbulent air impacts planes and their systems, leading to better designs for safety and performance. It’s a great example of how different areas of science work together to tackle big challenges in aviation. In here i was working under Dr Mrudula G in atmospheric electromagnetics.

2. Understanding Turbulence: Occurrence and Main Variables

Turbulence is when air or water moves in a messy, unpredictable way, like swirling winds or choppy waves. It happens when something, like fast-moving air, overcomes the forces that keep flow smooth, such as the thickness of the fluid. Things that affect turbulence include how fast the air is moving, how thick or sticky the fluid is, its density, and the size of whatever it’s flowing around—like an airplane wing. Knowing these basics helps people predict turbulence and manage it in everyday situations, like making flights safer.

3. Datasets (NetCDF4, Grid2): Challenges and Extraction Methods

NetCDF4 and Grid2 are formats used to store big sets of scientific data, like weather or turbulence info, but they’re tricky to read because they’re made for computers, not people. They pack in lots of details in a way that needs special tools to unlock. To simplify them, you can use something like Python to pull out the important bits and turn them into a CSV file, which is much easier to look at and use. This process makes complex data more approachable for anyone wanting to study it.

4. Using GrADS for Plotting NetCDF Data

GrADS, or Grid Analysis and Display System, is a handy tool for turning complex data—like the NetCDF files—into pictures like maps or graphs. It lets you see how things like wind speed or temperature change across places or times, which is perfect for spotting turbulence. By plotting this data, you can figure out where turbulence might be a problem and why. It's a practical way to visualize patterns and make sense of tricky information without needing to be an expert.

5. Converting Upper Air Sounding Data to CSV:

Upper air sounding data comes from weather balloons and tells us about the atmosphere—like temperature or pressure at different heights—but it's often stored in hard-to-read formats. To make it simpler, you can convert it into a CSV file, which is just a basic table you can open easily. This involves using a tool like Python to grab the key details and organize them into rows and columns. Once it's in CSV form, it's a lot easier to analyze and understand what's happening in the air.

6. Tracking Turbulence at Specific Times Using GrADS

With GrADS, you can zoom in on turbulence at a specific moment by picking a time in your data and plotting things like wind speed or pressure. This shows you exactly what's happening right then—like during a storm or a flight—and helps spot when turbulence pops up. It's a useful way to track how turbulence changes over time and understand its timing, which can be important for planning or safety.

7. Tracking Turbulence at Specific Locations Using GrADS

GrADS also lets you focus on turbulence in a specific place by entering its coordinates, like near an airport or over mountains. You can plot the data for that spot to see how turbulence behaves there, helping you find areas where it's more common or stronger. This is great for identifying trouble spots and understanding how the landscape affects air movement, making it easier to plan safer paths or study local weather patterns.

CHAPTER 5
CHALLENGES AND SOULTION

1. Dataset Selection

Challenges

The selection of an appropriate dataset was a pivotal decision in this study, with three options evaluated: the Climate Data Store (CDS), IMDAA, and Upper Air Sounding data. Each dataset presented distinct challenges:

- **Climate Data Store (CDS):** As a global dataset, CDS provides comprehensive coverage. However, it is constrained by request size limitations, which restrict the number of variables and timestamps that can be downloaded simultaneously.
- **IMDAA:** This Asia-level dataset offers data at 3-hourly intervals. While useful, this temporal resolution risks missing critical intervals that could be significant for analysis.
- **Upper Air Sounding:** Sourced from the Ogimet website, this dataset is confined to airport locations and includes METAR data. Its primary limitation lies in its restricted geographical scope and the need for format conversion to enhance usability.

Solutions

To address these challenges, tailored solutions were implemented for each dataset:

- **Climate Data Store (CDS):** Data retrieval was optimized by downloading it in weekly intervals, capturing all required variables in a single dataset, or in monthly intervals for individual variables. These were subsequently concatenated or merged into a CSV file for model training. Alternatively, the data was retained in its native NC file format, with variables extracted and stored separately for use.
- **IMDAA:** The 3-hourly data was utilized as provided, without interpolation or averaging of adjacent values. This approach avoided potential biases or data manipulation, preserving the integrity of the original dataset.
- **Upper Air Sounding:** Python scripts were developed to transform the data into a readable table format, making it accessible and comprehensible to a broader audience.

Conclusion

After careful consideration, the Climate Data Store (CDS) was selected for further processing. Its extensive range of variables, levels, and time selections offers significant advantages for future model training and testing, outweighing the limitations of the other datasets.

2. Model Selection Process

Challenges

Three models were assessed for their ability to predict turbulence, each encountering specific challenges:

- **LSTM + CNN (Hybrid Model):** This hybrid approach struggled to accurately capture turbulence intensity peaks, resulting in suboptimal predictive performance.
- **PINN with RANS:** This model leverages Physics-Informed Neural Networks (PINN) to solve the Reynolds-Averaged Navier-Stokes (RANS) equations, incorporating Reynolds stress relative to the mean velocity field. While theoretically robust, it proved highly complex and computationally demanding, requiring substantial resources such as dual RTX 5000 series cards.
- **Random Forest and XGBoost Hybrid Model:** This model offered faster training times compared to the alternatives. However, it faced the risk of overfitting or overlearning, potentially compromising its generalizability.

Solution

The Random Forest and XGBoost hybrid model are good for turbulence prediction. This choice was driven by its favourable balance of training efficiency and predictive accuracy. To mitigate the risk of overfitting, careful hyperparameter tuning and rigorous validation processes were employed.

HARDWARE AND SOFTWARE REQUIREMENTS

Software requirements:

- **Operating System:**
    - Preferred: Windows 10/11 or Ubuntu 20.04+

A stable operating system is necessary to run Python environments and support visualization libraries without compatibility issues. Ubuntu is especially suitable for scientific computing.

- **Programming Language:**
    - Python 3.8 or higher

Python served as the primary language for data processing, analysis, and visualization. Compatibility with libraries such as Pandas, NumPy, and Matplotlib was ensured through the latest Python version.

- **Development Environment:**
    - Jupyter Notebook or VS Code (Visual Studio Code)

Used for writing, testing, and documenting Python scripts. Jupyter facilitated interactive data analysis, while VS Code was used for more structured development and debugging.

- **Required Python Libraries/Packages:**
    - Pandas – for data manipulation and analysis
    - NumPy – for numerical operations and array handling
    - Matplotlib/Seaborn – for plotting and visualizing data such as Vwind graphs
    - All libraries were installed and managed using pip or conda.

- **Data Sources/Formats:**
    - Support for data formats such as .csv, .txt, or .json, based on upper-air sounding datasets.
    - Optional: Integration with atmospheric data APIs or datasets from NOAA or Indian Meteorological Department (IMD) and ERA5 Data from Climate Data Store (CDS)

Hardware Requirements:

- **PROCESSOR:**
    - Minimum: Intel Core i5 (8th Gen or above) / AMD Ryzen 5
    - A multi-core processor was needed to handle data parsing and plotting without performance lag.

- **RAM:**
    - Minimum: 8 GB
    - Recommended: 16 GB

  Sufficient memory was essential for handling large meteorological datasets, especially when using in-memory data frames with Pandas.

- **STORAGE:**
    - Minimum: 256 GB
    - Recommended: 512 GB

  Fast storage was important for quick read/write access to datasets and improved overall system performance during data-intensive operations.

- **DISPLAY AND GPU:**
    - A high-resolution display helped in better visualization and interpretation of plots.
    - GPU was not mandatory since the project did not involve deep learning or graphical rendering, but optional for future extensions involving ML models.

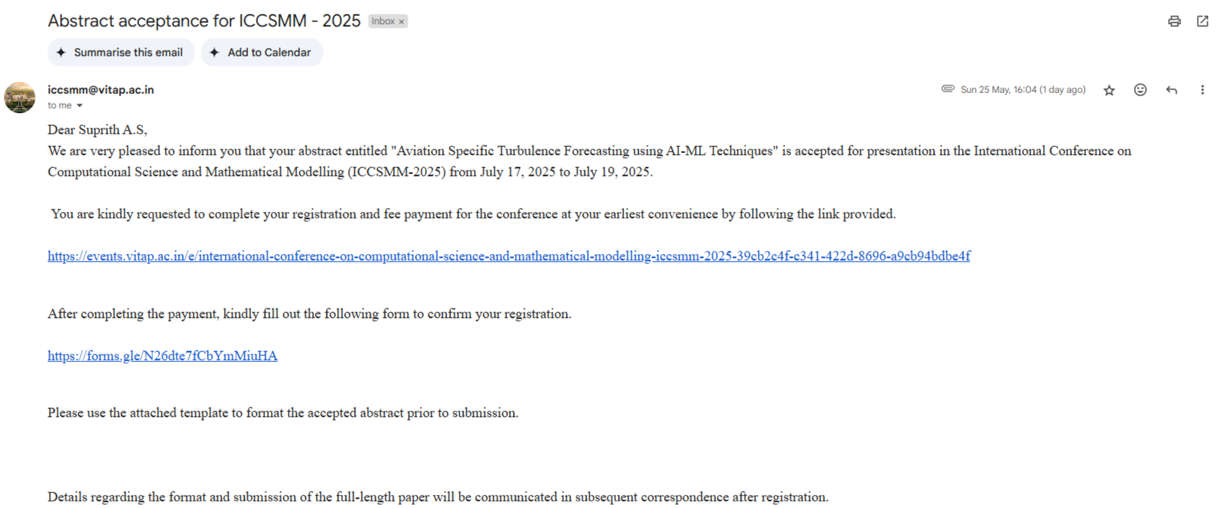
- **INTERNET CONNECTIVITY:**
    - Required for downloading data, installing libraries, and accessing online documentation or APIs. Stable internet ensured uninterrupted workflow.

CHAPTER 6

ACHEIVEMENTS AND CONTRIBUTION

ACHIEVEMENTS:

One of the standout achievements of my internship was the acceptance of my abstract, titled "Aviation Specific Turbulence Forecasting using AI-ML Techniques", for presentation at the International Conference on Computational Science and Mathematical Modelling (ICCSMM-2025), scheduled to take place from July 17 to July 19, 2025. This accomplishment highlights the quality and importance of the research I conducted, which explored the use of artificial intelligence and machine learning to enhance turbulence forecasting in aviation—an area with significant implications for improving flight safety and operational efficiency. Having my work recognized at a prestigious international conference reflects the technical expertise, problem-solving abilities, and innovative thinking I developed during the internship. Through this project, I strengthened my skills in research design, data analysis, and effective communication of complex ideas, all of which were critical to producing a submission deemed worthy of a global stage. This experience not only marks a meaningful contribution to the field of computational science but also equips me with valuable insights and capabilities for my future academic and professional journey.


Abstract acceptance for ICCSMM - 2025 Inbox ×

Summarise this email | Add to Calendar

iccsmm@vitap.ac.in
to me

Sun 25 May, 16:04 (1 day ago)

Dear Suprith A.S,
We are very pleased to inform you that your abstract entitled "Aviation Specific Turbulence Forecasting using AI-ML Techniques" is accepted for presentation in the International Conference on Computational Science and Mathematical Modelling (ICCSMM-2025) from July 17, 2025 to July 19, 2025.

You are kindly requested to complete your registration and fee payment for the conference at your earliest convenience by following the link provided.

https://events.vitap.ac.in/e/international-conference-on-computational-science-and-mathematical-modelling-iccsmm-2025-39cb2c4f-c341-422d-8696-a9cb94bdbe4f

After completing the payment, kindly fill out the following form to confirm your registration.

https://forms.gle/N26dte7fCbYmMiuHA

Please use the attached template to format the accepted abstract prior to submission.

Details regarding the format and submission of the full-length paper will be communicated in subsequent correspondence after registration.

Figure 6.1: Abstract acceptance mail for ICCSMM

CONTRIBUTION:

Understanding Turbulence with GrADS

Turbulence in the atmosphere refers to chaotic air motions that can affect aviation, such as clear-air turbulence (CAT). GrADS is ideal for studying turbulence because it can handle gridded data (e.g., model outputs) and point data (e.g., upper air soundings), allowing you to visualize and analyze conditions that lead to turbulence, like wind shear or temperature inversions.

- **Why GrADS?**

  GrADS excels at plotting meteorological fields such as wind, temperature, and pressure, which are key to identifying turbulence. It supports data visualization over specific locations (using latitude and longitude) and can overlay this data on maps, such as those showing airports.

Airport Images and Latitude/Longitude

GrADS can generate maps with meteorological data overlaid on airport locations, specified by latitude and longitude. This is particularly helpful for visualizing turbulence near airports, which is critical for aviation safety.

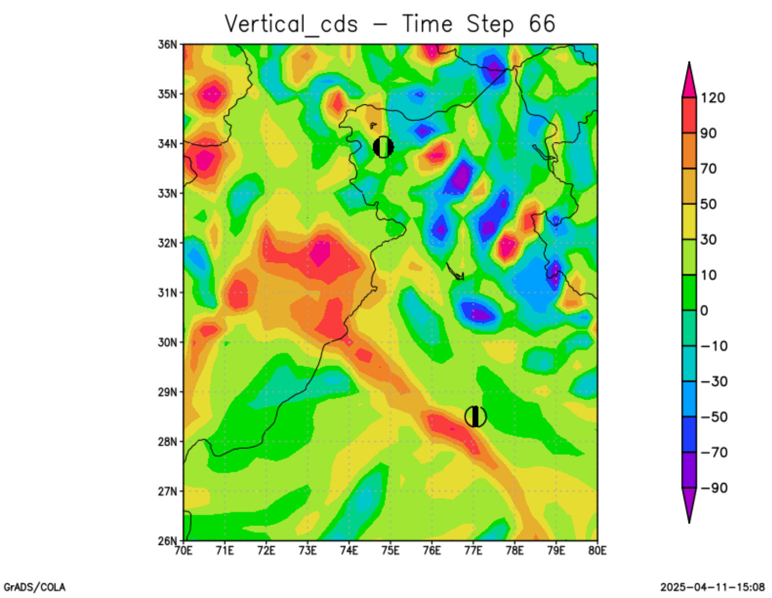


Figure 6.2: GrADS on incident 19th Feb 2024 Turbulence on flight route Delhi to Srinagar

Normal GrADS Data vs. CSV Data

GrADS handles different data formats, which is relevant when working with turbulence data from various sources.

- **Normal GrADS Data**:
  - Typically binary grid files (e.g., GRIB, NetCDF) used for model outputs or reanalysis data. GrADS reads these using control files (.ctl).
  - Example: Open a GRIB file with a .ctl file and plot its contents directly.
- **CSV Data**:
  - Point data, like upper air soundings, might come in CSV format. To use it in GrADS, you must convert it to a binary or ASCII grid file using external tools or scripts.
- **Comparison**:
  - Binary grids are efficient for large datasets (e.g., model results), while CSV is better for smaller, point-based data (e.g., soundings). For turbulence, you might use CSV for specific soundings and binary grids for broader forecasts.

Upper Air Sounding Data and Turbulence

Upper air soundings, collected by weather balloons, provide vertical profiles of temperature, humidity, wind speed, and direction. These profiles are critical for understanding turbulence because they reveal atmospheric stability and wind shear—conditions often linked to turbulent air.

- **How It Works**:
  - Soundings can show layers where wind speed changes sharply with height (wind shear), a common cause of turbulence due to phenomena like Kelvin-Helmholtz instability.
  - Stability indicators, such as the Richardson number (a ratio of buoyancy to shear), can also be derived from sounding data to pinpoint turbulence-prone regions.

The dataset in question consists of meteorological observations collected from a weather station in Bangalore, India (station code 43295), covering the month of March 2025. It includes critical atmospheric parameters such as pressure, temperature, dew point, wind direction, and wind speed, recorded at various altitudes and time intervals. The data is presented in two distinct formats: a detailed report encoded in standard meteorological formats like PPCC and TTAA, which provides comprehensive weather information across multiple atmospheric levels from March 1 to March 31, 2025, and a structured spreadsheet excerpt from March 1, 2025, offering a tabular view of the same parameters for that specific day. This dataset is invaluable for atmospheric modeling, weather forecasting, and climate research, as it enables the analysis of weather patterns, the development of predictive models, and the study of climate variability in the region.

Query made at 04/11/2025 19:12:09 UTC

Time interval: from 03/01/2025 00:00 to 03/31/2025 23:00 UTC

43295, Bangalore (India)
ICAO index: ----. **WIGOS Id:** 0-20000-0-43295
Latitude 12-58-00N. **Longitude** 077-34-59E. **Altitude** 917 **m.**

TEMP/PILOT from 43295, Bangalore (India)

PPCC 01/03/2025 00:00->	**PPCC 51002 43295 55370 29020 25526 26024 77999=**
TTAA 01/03/2025 00:00->	**TTAA 51001 43295 99910 20024 09006 00/// ///// ///// 92/// ///// ///// 85508 18856 09517 70148 11892 08015 50587 03769 06515 40759 17169 08516 30971 28579 30520 25098 39557 24021 20247 52765 26027 15426 67161 28021 10663 78361 32013 88999 77999 31313 53808 82301=**
TTBB 01/03/2025 00:00->	**TTBB 51008 43295 00910 20024 11873 19034 22855 18856 33771 14059 44738 12064 55730 12470 66719 12679 77678 10294 88644 08283 99559 01280 11535 00572 22499 03769 33372 20769 44366 20972 55363 20376 66360 19779 77357 19181 88331 23180 99266 36164 11255 38959 22228 45164 33161 64161 44132 73561 55105 78161 21212 00910 09006 11874 10018 22373 05008 33346 34030 44268 27516 55108 32021 66105 32519 31313 53808 82301 41414 20901 51515 11900 08011 22800 11511 33600 09504=**
TTCC 01/03/2025 00:00->	**TTCC 51002 43295 70865 75762 29020 50062 66967 25526 30378 60980 26024 20634 51383 08009 88879 82361 25516 77999 31313 53808 82301=**
TTDD 01/03/2025 00:00->	**TTDD 5100/ 43295 11879 82361 22782 78362 33756 75962 44506 67767 55428 62979 66334 61580 77272 60580 88231 55382 99155 47984 11113 45785 21212 11951 27507 22347 25537 33207 15001 44113 09056 31313 53808 82301=**
PPAA 01/03/2025 06:00->	**PPAA 5106/ 43295 NIL=**
PPBB 01/03/2025 06:00->	**PPBB 5106/ 43295 NIL=**
PPCC 01/03/2025 06:00->	**PPCC 5106/ 43295 NIL=**
PPDD 01/03/2025 06:00->	**PPDD 5106/ 43295 NIL=**
PPCC 01/03/2025 12:00->	**PPCC 5112/ 43295 NIL=**
TTAA 01/03/2025 12:00->	**TTAA 5112/ 43295 NIL=**
TTBB 01/03/2025 12:00->	**TTBB 5112/ 43295 NIL=**
TTCC 01/03/2025 12:00->	**TTCC 5112/ 43295 NIL=**
TTDD 01/03/2025 12:00->	**TTDD 5112/ 43295 NIL=**
PPCC 01/03/2025 18:00->	**PPCC 5118/ 43295 NIL=**
TTAA 02/03/2025 00:00->	**TTAA 52001 43295 99908 22040 09006 00/// ///// ///// 92/// ///// ///// 85493 21261 08012 70140 11277 07015 50587 04183 01522 40759 17173 10511 30970 30175 27001 25097 41360 19514 20245 51571 20521 15425 66764 29013 10663 79969 12505 88999 77999 31313 53808 82301=**

Fig 6.3: Original Upper Air Sounding data.

	A	B	C	D	E	F	G	H	I
1	Timestamp	MessageType	StationInfo	Pressure_hPa	Temperature_C	DewPoint_C	WindDirection_deg	WindSpeed_kt	
2	3/1/2025	PPCC	51002 43295	370	29	27	255	26	
3	3/1/2025	PPCC	51002 43295	24	77.9	68			
4	3/1/2025	TTAA	51001 43295	910	20	17.6	90	6	
5	3/1/2025	TTAA	51001 43295	508	18.8	13.2	95	17	
6	3/1/2025	TTAA	51001 43295	148	11.8	2.6	80	15	
7	3/1/2025	TTAA	51001 43295	587	3.7	-3.2	65	15	
8	3/1/2025	TTAA	51001 43295	759	17.1	10.2	85	16	
9	3/1/2025	TTAA	51001 43295	971	28.5	20.6	305	20	
10	3/1/2025	TTAA	51001 43295	98	39.5	33.8	240	21	
11	3/1/2025	TTAA	51001 43295	247	52.7	46.2	260	27	

Fig 6.4: CSV file after extraction of data from the Upper Air Sounding data.

CHAPTER 7

REFLECTION AND ANALYSIS

Understanding of the turbulence from the aave graphs:

We analyzed the Aave graphs to pinpoint the periods of highest turbulence.

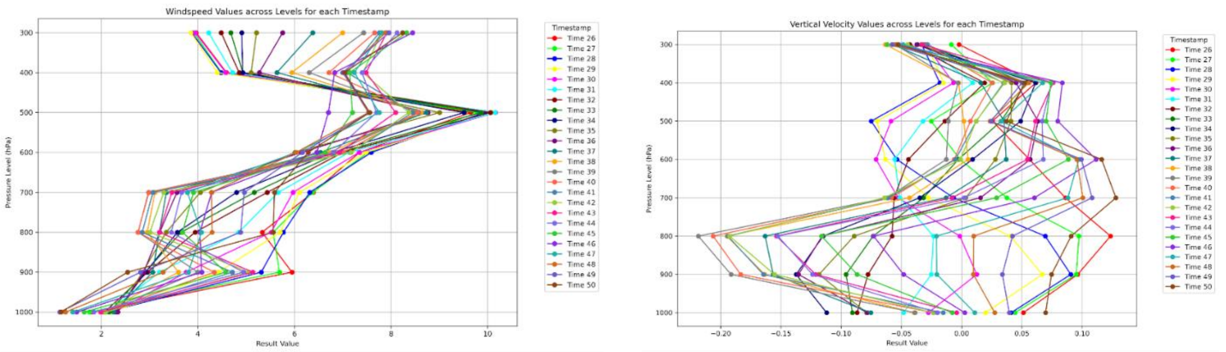


Figure 7.1: Windspeed(wspd), Vertical Velocity(w)

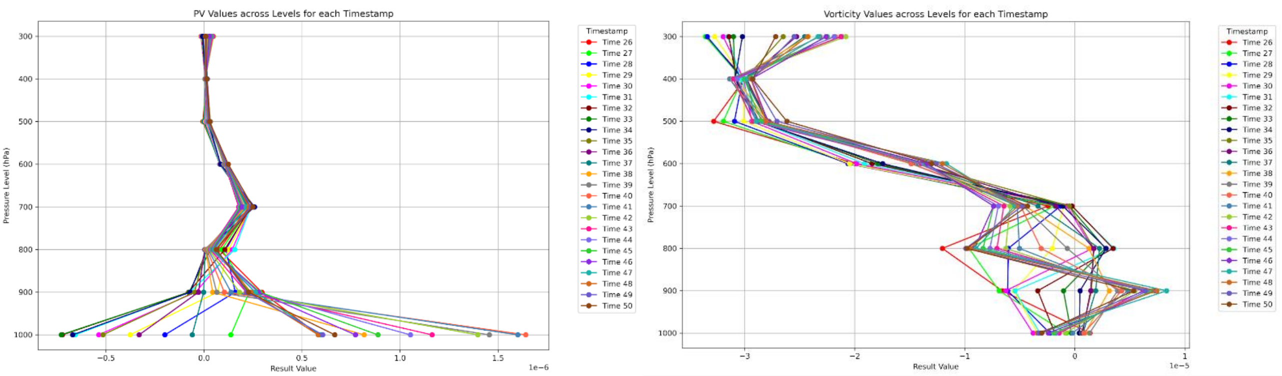


Figure 7.2: Potential Vorticity(pv), Vorticity Relative(vo)

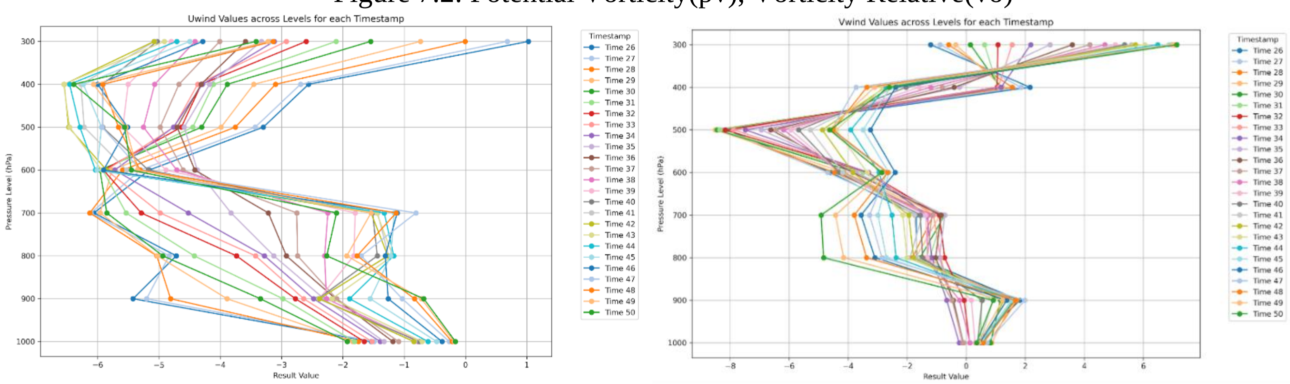


Figure 7.3: Uwind (u), Vwind(v)

The above graphs are of the incident of Flight IndiGo (6E498) traveling from Jaipur to Bengaluru on 2nd March 2025.In the U-wind component shows a sharp fall at 1000 and 900 levels between Timestamps 26 and 30, where the U-wind speed becomes less negative. Another significant change is a sharp rise at multiple levels between Timestamps 37 and 38. These periods, specifically around Timestamps 29-30 and 37-38, are identified as potentially turbulent.

The V-wind component exhibits a sharp decrease at 900 and 800 levels from Timestamps 32 to 34, where the V component rapidly becomes negative. A steep fall in V at multiple levels is also observed between Timestamps 40 and 47. Consequently, the periods around Timestamps 32-34 and 40-47 are highlighted as areas of possible turbulence for the V-wind.

Further insights are gained from the Wind Speed graph, which displays rapid changes across all levels between Timestamps 26 and 30. More fluctuations are evident from Timestamps 40 to 47. These observations suggest possible turbulence around Timestamps 26-30 and 40-47 for overall wind speed.

The Relative Vorticity also shows a large drop in vertical vorticity at multiple levels between Timestamps 26 and 30, followed by a continued sharp decrease, particularly at 800 and 700 levels, from Timestamps 40 to 47.

These periods align with potential turbulence. The Vertical Velocity component demonstrates a sharp drop in vertical wind (W component) at multiple levels between Timestamps 29 and 32. A recovery, marked by a positive rise after negative values, is noted from Timestamps 47 to 50. The period around Timestamps 29-32 is considered potentially turbulent for vertical velocity.

Lastly, the Potential Vorticity graph indicates a sharp drop at the 1000 level between Timestamps 26 and 30, and a sudden jump at the 1000 level (from negative to high positive) between Timestamps 37 and 38. The latter period, around Timestamps 37-38, is identified as a likely area of turbulence. From these above observations we come up with three distinct turbulence-prone areas that are identified based on the observed changes across these parameters.

The first event is likely around Timestamps 29-30, characterized by an initial drop in U-wind, Potential Vorticity, Relative Vorticity, and Vertical Velocity. The second event occurs around Timestamps 37-38, marked by a recovery in Potential Vorticity and U-wind. The third event, between Timestamps 40 and 47, is associated with sharp fluctuations in V-wind, Relative Vorticity, and overall Wind Speed.

GrADS (Grid Analysis and Display System) to analyze relative vorticity along the flight route from Jaipur to Bengaluru on March 2nd, 2025, would provide crucial insights into atmospheric conditions impacting the flight. This analysis specifically aims to identify the pressure levels most affected by significant vorticity. By examining the vertical profile of relative vorticity, areas of enhanced rotation or shear in the atmosphere can be pinpointed, which are directly correlated with potential turbulence.

Such an investigation would involve visualizing the vorticity fields at different pressure levels (e.g., 1000hPa, 900hPa, 800hPa, etc.) along the flight path. The goal is to identify at which specific pressure levels the relative vorticity values show sudden changes or significant magnitudes, indicating where turbulent eddies or strong wind shear might be present. This targeted approach helps in understanding the vertical extent and intensity of any atmospheric disturbances, allowing for more precise forecasting of turbulence and potential adjustments to flight altitudes for passenger comfort and safety.

GrADS for variable Vorticity (Relative) incident flight route Jaipur to Bengaluru.

Date: 02/03/2025, 2nd March 2025.

(this was mainly used to find the exact pressure level it had effected the most.):

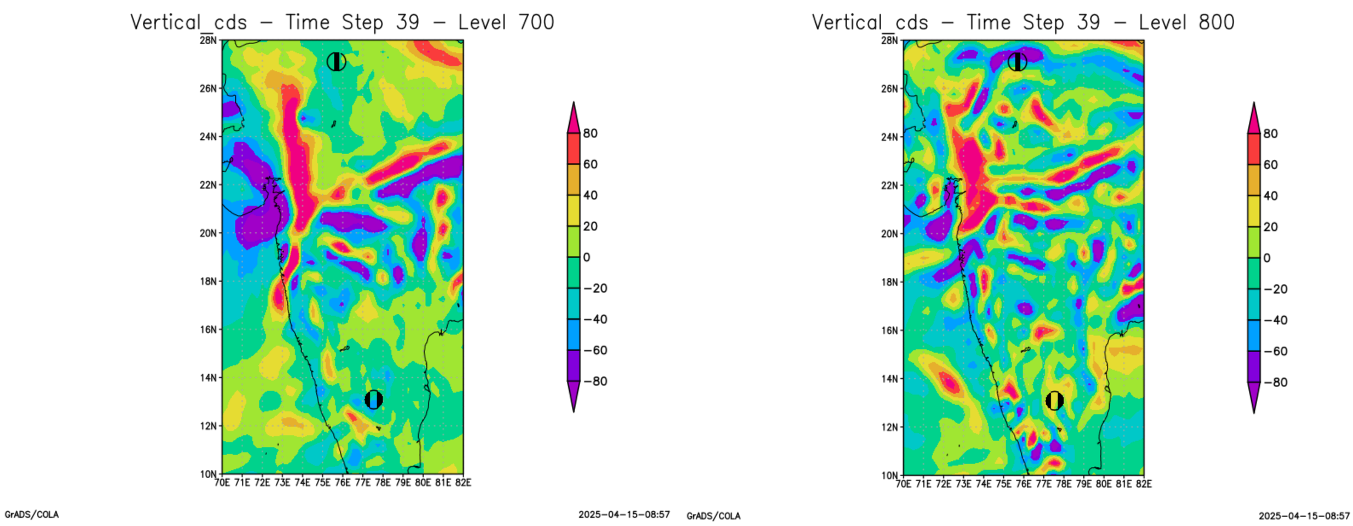


Figure 7.4: Vorticity (Relative) GrADS at level 700 and 800

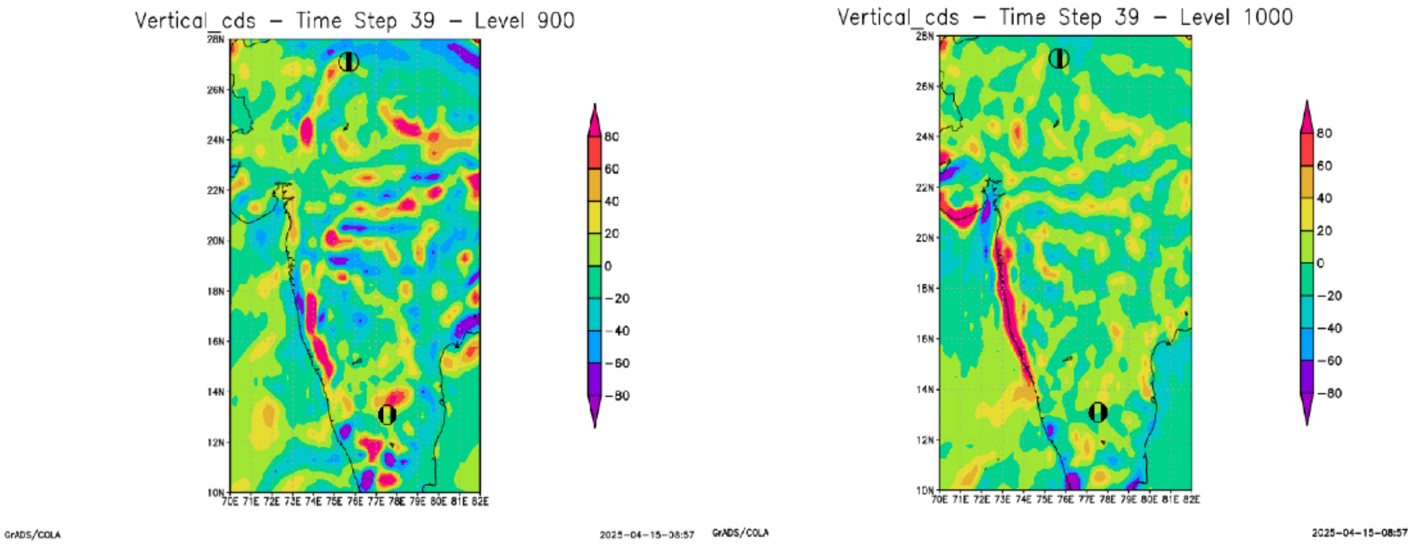


Figure 7.5: Vorticity (Relative) GrADS at level 900 and 1000

On February 19, 2024, the flight route from Delhi to Srinagar experienced significant atmospheric disturbances, which could be thoroughly investigated using GrADS to visualize relative vorticity and vertical velocity. Reports from that date confirm that an IndiGo flight (6E6125) on this route encountered severe turbulence due to inclement weather, with passengers reporting a harrowing experience. This suggests a strong presence of both relative vorticity (indicating rotational motion and shear in the atmosphere) and significant vertical velocities (updrafts and downdrafts), Which are direct causes of turbulence. A GrADS analysis for this specific date would reveal the precise pressure levels where high magnitudes of relative vorticity and vertical velocity were concentrated. This would likely correspond to the presence of active Western Disturbances affecting the Western Himalayas, bringing widespread rain and snowfall to Jammu and Kashmir, as predicted and observed on that day. Identifying these turbulent layers would be crucial for understanding the meteorological mechanisms behind the reported flight disruptions and could inform future flight planning and safety protocols for this frequently affected route.

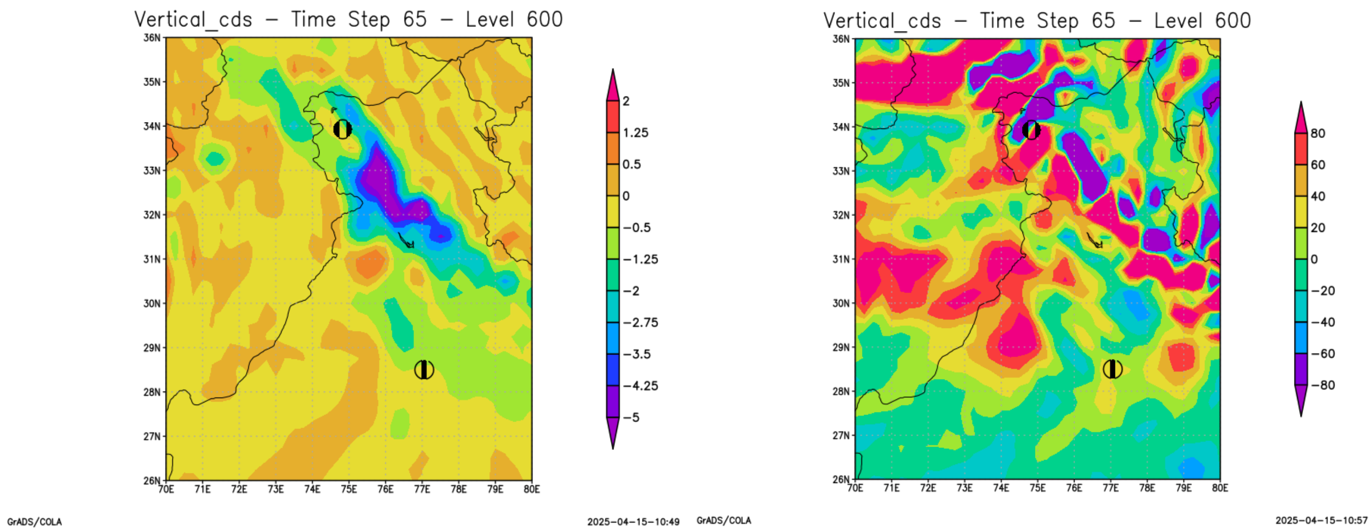


Figure 7.6: Vertical Velocity (600 hpa) And Vorticity_Relative (600 hpa)

GrADS helps locate turbulence by plotting variable vertical velocity across **latitude, longitude, and atmospheric levels**. This creates a 2D view, precisely identifying where and at what altitude turbulence occurred along a flight path.

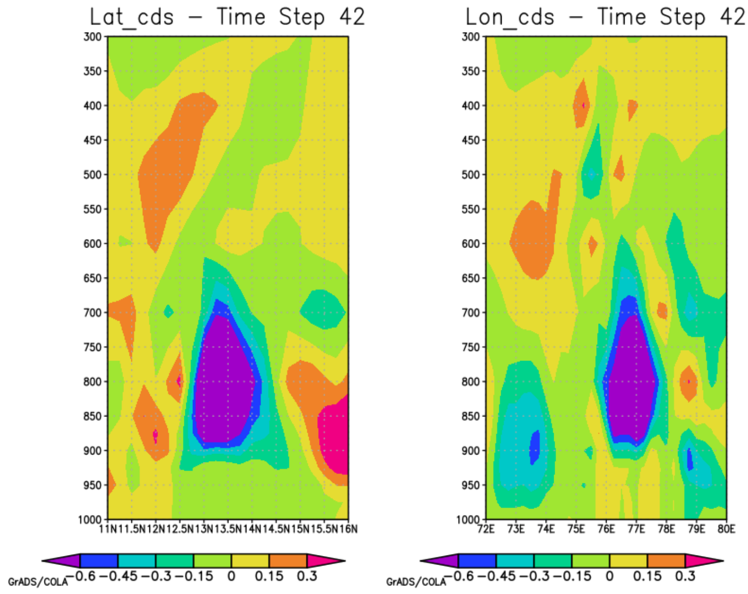


Figure 7.7: latitude and longitude vs level using GrADS

GrADS helps find turbulence by visualizing **wind shear**—sudden changes in wind direction or speed. Areas with rapidly shifting wind vectors or converging/diverging streamlines indicate potential turbulence.

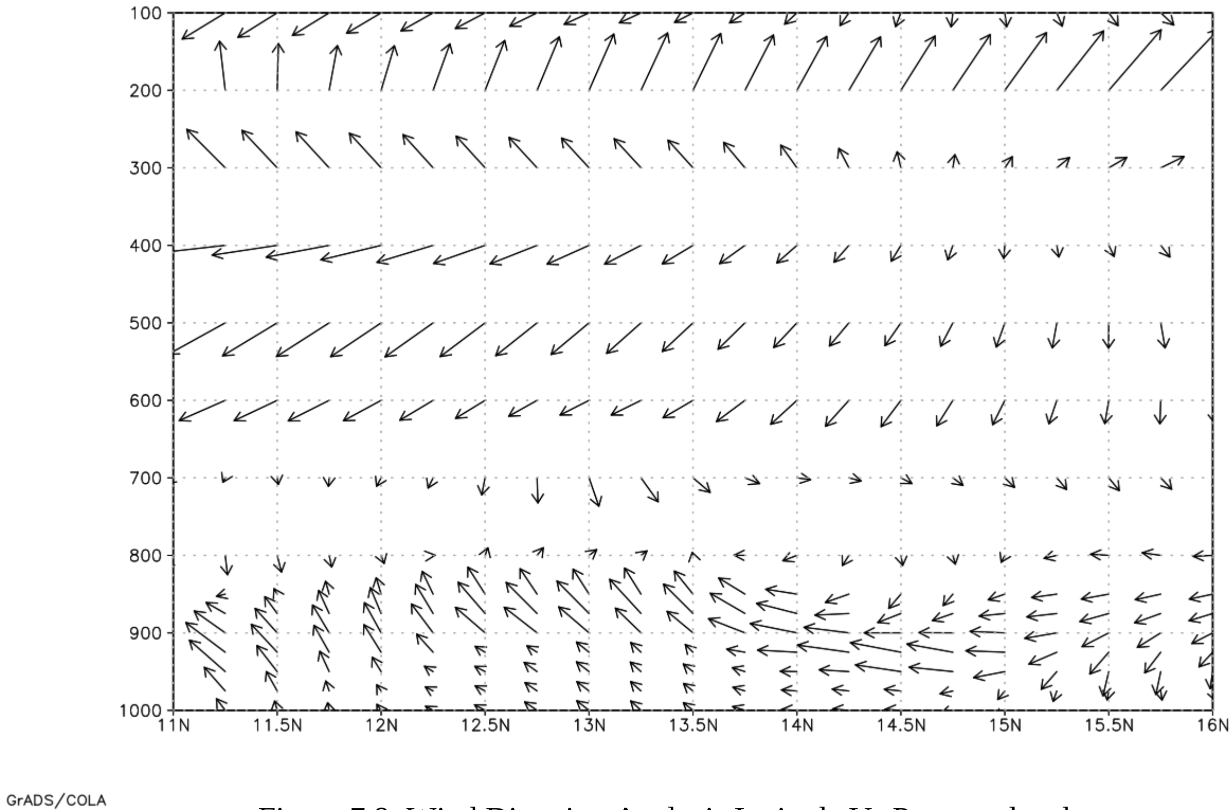


Figure 7.8: Wind Direction Analysis Latitude Vs Pressure level

This is a vector field plot depicting wind patterns over a geographical region, In the Southern Hemisphere, as indicated by the latitude range from 11N to 16N and longitude from 75E to 85E. The arrows represent wind direction and magnitude, with their length and orientation showing the strength and flow of the wind at various points. At higher altitudes (100 to 400 units on the y-axis), the winds predominantly blow from the southwest to the northeast, suggesting a consistent westerly flow. As the altitude decreases (500 to 1000 units), the wind patterns become more chaotic, with directions varying significantly, indicating turbulence or shifting weather systems.

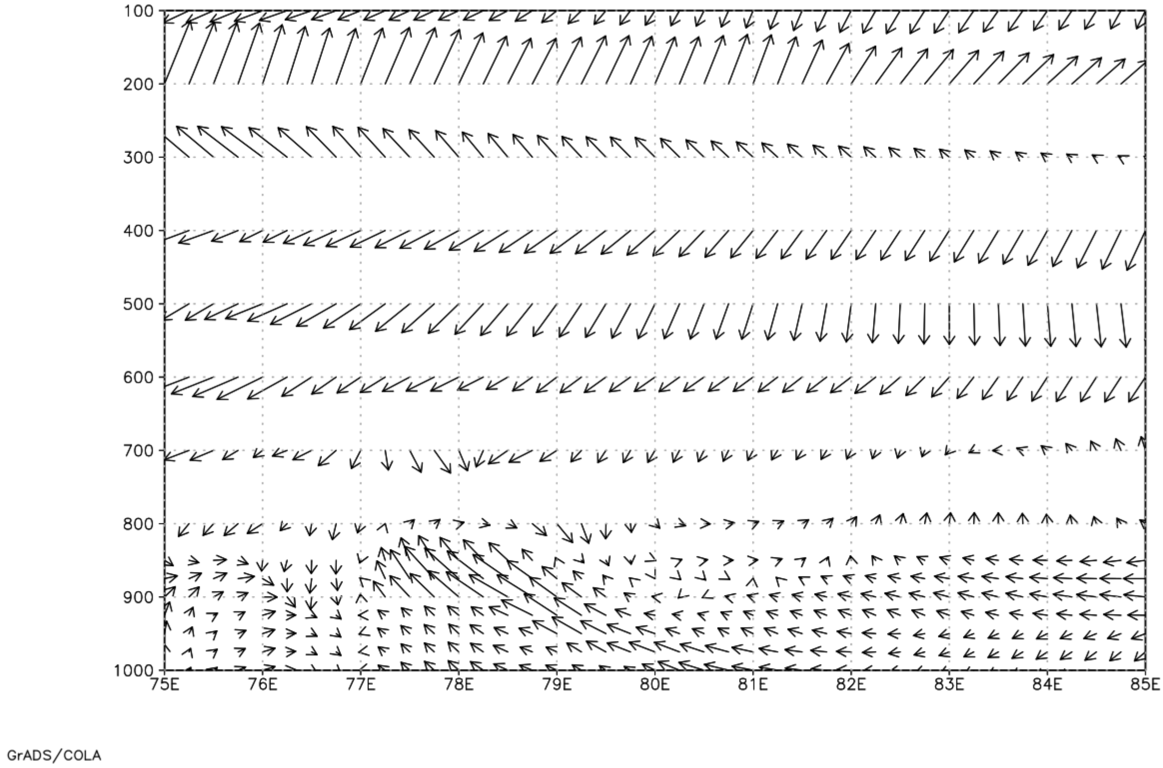


Figure 7.9: Wind Direction Analysis Longitude Vs Pressure level

This is a vector field plot, similar to the first, but covering a latitude range (11N to 16N) and the same longitude range (75E to 85E). Like the first, it shows wind patterns with arrows indicating direction and magnitude. At higher altitudes (100 to 400 units), the winds are mostly uniform, blowing from the southwest to the northeast, consistent with a westerly flow pattern. However, at lower altitudes (500 to 1000 units), the wind directions become more varied, with some areas showing winds blowing in opposing directions, suggesting complex weather dynamics or a transition zone.

CHAPTER 8

RECOMMENDATIONS

The prediction and forecasting of atmospheric turbulence remain a significant challenge in aviation meteorology, directly impacting flight safety, passenger comfort, and operational efficiency. While traditional numerical weather prediction (NWP) models provide a foundational understanding of atmospheric dynamics, their ability to resolve the small-scale and transient nature of turbulence is often limited. To bridge this gap and enhance predictive capabilities, the adoption of advanced machine learning (ML) and artificial intelligence (AI) models is highly recommended. These models offer the potential to learn complex, non-linear patterns from vast atmospheric datasets, leading to more accurate and timely turbulence forecasts, ultimately contributing to safer and more efficient flight operations.

One promising approach for turbulence forecasting involves the use of a hybrid Long Short-Term Memory (LSTM) and Convolutional Neural Network (CNN) architecture. This model is particularly well-suited for handling the spatio-temporal nature of atmospheric data. CNNs can effectively extract spatial features from gridded data, such as temperature and wind fields, identifying patterns indicative of shear or localized atmospheric instabilities. The LSTM component can then process these features sequentially, capturing temporal dependencies and learning how atmospheric conditions evolve over time to produce turbulence. This hybrid model can be trained to predict turbulence intensity or probability at specific locations and future time instances, offering a significant improvement over existing diagnostic tools by providing more nuanced and actionable intelligence.

Physics-Informed Neural Networks (PINNs) integrated with Reynolds-Averaged Navier-Stokes (RANS) equations present a cutting-edge solution. PINNs embed physical laws directly into the neural network's learning process, ensuring that predictions are consistent with fundamental fluid dynamics principles. In the context of turbulence, a PINN could learn the closure models for RANS equations directly from observational and NWP data, while being constrained by the physics of fluid motion. This method could be particularly beneficial for resolving sub-grid scale turbulence, which is often parameterized in traditional models, and for improving the representation of turbulent kinetic energy and Reynolds stresses. While computationally intensive, the potential for more accurate and physically consistent turbulence forecasts makes PINNs a valuable area for future research and development.

Ensemble learning algorithms like Random Forest (RF) and Extreme Gradient Boosting (XGBoost) offer robust and often highly accurate alternatives, particularly when computational resources are a consideration or when rapid prototyping is needed. These models are adept at handling high-dimensional datasets and complex, non-linear relationships between atmospheric variables and turbulence occurrence. By engineering relevant features from variables such as temperature, u-component of wind, v-component of wind, vertical velocity, and relative vorticity (e.g., calculating various shear indices, Richardson number, or other stability parameters), RF and XGBoost models can be effectively trained to classify turbulence intensity levels or predict continuous turbulence metrics. Their inherent ability to provide feature importance can also offer valuable insights into the key drivers of turbulence in different atmospheric regimes, aiding in a better physical understanding. The adoption of these AI/ML models for turbulence prediction offers several key benefits that can transform aviation meteorology. Firstly, they can significantly improve the accuracy and lead time of turbulence forecasts by learning from vast amounts of historical data and identifying subtle precursor patterns that traditional methods might miss. This directly leads to enhanced flight safety by allowing pilots and air traffic controllers to make more informed decisions, such as adjusting flight paths or altitudes to avoid or mitigate encounters with hazardous turbulence. More precise turbulence forecasts can contribute to substantial operational efficiencies, including optimized flight routing to reduce fuel consumption, minimize delays, and enhance passenger comfort.

To maximize the effectiveness of these advanced models, it is crucial to integrate diverse and high-quality datasets. This includes high-resolution NWP outputs, pilot reports (PIREPs), data from airborne sensors, satellite imagery, ground-based radar data, and upper air soundings. The development of robust data assimilation techniques and feature engineering strategies will be key to unlocking the full potential of these datasets. Future work should also focus on developing probabilistic forecasts that quantify the uncertainty associated with turbulence predictions, providing decision-makers with a more complete picture of the potential risks. Collaborative efforts between research institutions, meteorological services, and the aviation industry will be essential to advance these technologies and translate them into operational tools that benefit the entire aviation ecosystem.

CHAPTER 9

CONCLUSION

This internship project, centered on the theme of "Aviation Specific Turbulence Forecasting using AI-ML Techniques," embarked on a critical mission to enhance flight safety and operational efficiency within the aviation sector. The core objective was to explore and develop innovative methodologies for turbulence detection and prediction by harnessing the power of Artificial Intelligence and Machine Learning. Turbulence, a complex and often unpredictable atmospheric phenomenon, poses significant risks to aircraft integrity, passenger safety, and crew well-being. Therefore, advancing the capability to accurately forecast turbulent conditions is of paramount importance to the aerospace industry, and this project represented a dedicated effort towards achieving this vital goal through modern computational approaches.

The project's methodological approach was rooted in the systematic integration and meticulous analysis of diverse atmospheric datasets. A significant part of the work involved identifying key meteorological parameters and variables that are precursors to or indicative of turbulent conditions, such as significant wind shear (both vertical and horizontal), strong temperature gradients, pronounced vertical air movements (vertical velocity), and areas of high vorticity. Data sources like upper air soundings, which provide detailed vertical profiles of the atmosphere, alongside comprehensive global reanalysis datasets such as ERA5 from the Climate Data Store (CDS), formed the foundational data backbone. Sophisticated Python-based tools and libraries were extensively employed for essential tasks including data extraction, rigorous processing, necessary transformations (e.g., converting complex scientific formats like NetCDF and Grid2 into more universally accessible CSV files), and insightful visualization using tools like GrADS, which facilitated initial exploratory analysis and understanding of turbulence-prone atmospheric states.

A central tenet of this research was the application of state-of-the-art AI/ML algorithms, often non-linear patterns and complex correlations within the voluminous atmospheric data that might signify the onset or presence of turbulence. The project involved an in-depth exploration and comparative assessment of various predictive models. These included hybrid CNN-LSTM networks, designed to leverage both spatial feature extraction capabilities of CNNs and temporal sequence learning of LSTMs and robust ensemble methods like Random Forest and XGBoost, known for their strong performance in classification and regression tasks on structured data. This exploration underscored the immense potential of AI/ML to move beyond traditional diagnostic indices towards more adaptive, data-driven, and potentially more accurate predictive systems for a phenomenon as multifaceted and dynamically evolving as atmospheric turbulence.

The internship was conducted in close collaboration with CSIR-National Aerospace Laboratories (NAL), a premier Indian R&D institution known for its pivotal contributions to aerospace science and technology. This partnership created an enriching environment where academic pursuits aligned seamlessly with real-world aerospace challenges. Working at CSIR-NAL's Centre for Electromagnetics (CEM) provided hands-on exposure to cutting-edge research in atmospheric sciences and their significant influence on aerospace systems and operations. This grounding ensured the project's objectives remained both scientifically rigorous and practically relevant to the aviation sector.

Key learnings included a deep understanding of atmospheric turbulence, the complexities of handling meteorological data, and applying advanced computational methods to solve real-world problems. The project involved evaluating meteorological datasets for turbulence research, building streamlined data preprocessing workflows, and comparing the performance of various AI/ML models for turbulence prediction. The acceptance of the research abstract, "Aviation Specific Turbulence Forecasting using AI-ML Techniques," at ICCSMM-2025 emphasize the research's value and relevance in the growing field of AI-driven meteorology.

In conclusion, this project established a strong foundation for applying AI/ML to improve the accuracy and reliability of turbulence forecasting. Challenges remain, especially regarding access to high-resolution, real-time turbulence data and operational deployment of models. Future work should focus on advanced model refinement, integrating diverse data sources, and validating results with real-world flight data. With continued efforts, such innovations could greatly enhance aviation safety, reduce weather-related disruptions, and contribute to a more efficient global air transport system.

CHAPTER 10
REFERENCES

1. "CSIR-National Aerospace Laboratories (CSIR-NAL), Bengaluru", India Science, Technology & Innovation Portal, Ministry of Science and Technology, Government of India, 2025.

2. "https://www.nal.res.in/en/about-us", CSIR-National Aerospace Laboratories, Bengaluru, 2025.

3. Chrit, M., Majdi, M., 'Operational wind and turbulence nowcasting capability for advanced air mobility', Neural Computing and Applications, Vol. 36, pp. 10637–10654, 2024.

4. Wang, J.-X., Wu, J.-L., Ling, J., Iaccarino, G., Xiao, H., 'Physics-Informed Machine Learning for Predictive Turbulence Modeling: Towards a Complete Framework', Center for Turbulence Research Proceedings of the Summer Program 2016, Stanford University, 2016.

5. Kakljugin, Peter, 'ML for Turbulence Prediction: How AI and ML are Revolutionizing Forecasting', Medium, 2023.

6. 'Deep Learning Approaches for Turbulence Modeling', arXiv:2307.13784, 2023.

7. 'Numerical Simulation of Turbulent Flows', Computation, MDPI, Vol. 12, Issue 1, Article 4, 2024.

8. 'Advancements in CFD for Aerospace Applications', Fluids, MDPI, Vol. 59, Issue 1, Article 24, 2024.

9. "IndiGo Flight Performs Go-Around at Bengaluru", Aviation A2Z, March 4, 2025.

10. "Video: Passengers Terrified After IndiGo Flight Hit By Turbulence", NDTV, On February 19th 2024.